

Workflow Solution for ISO
Document Library
SRS
Version II

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Introduction

Purpose

The purpose of this system is to automate the document upload, validation, version control, approval workflow, and publishing process. It ensures that only valid documents are approved and published while maintaining document version history.

Scope

The system allows users to:

- Upload documents.
- Validate document format (ISO format).
- Detect duplicate documents.
- Compare document versions.
- Create new versions when changes are detected.
- Determine approver according to approval matrix
- Route documents for approval.
- Notify users of approval or rejection

Audience

- GPSL employee
- GPSL management(Approver/Reviewer)
- Administrator

Definition

- ISO - International Organization for standerlization

Overall Description

Product Perspective

The system replaces manual document handling by providing an automated workflow that validates uploaded documents against ISO standards, detects duplicate documents, compares document versions, identifies modifications, manages document versioning, routes documents to the appropriate approver based on an approval matrix, and publishes approved documents to SharePoint.

Product Functions

- Upload ISO-compliant documents.
- Validate uploaded document format.
- Display error messages for invalid documents.
- Detect duplicate documents.
- Compare uploaded documents with existing versions.
- Identify added, removed, or modified sections.
- Automatically create new document versions.
- Notify users when identical documents already exist.
- Determine the appropriate approver using an approval matrix.
- Send approval requests to designated approvers.
- Allow approvers to approve or reject documents.
- Notify users of approval or rejection decisions.
- Publish approved documents to SharePoint.

User Classes and Characteristics

- Document Creator (Primary User) - Employees responsible for creating and uploading documents.
- Approver/Reviewer - Managers or department heads responsible for reviewing documents.
- System Administrator - Technical personnel responsible for managing the system.

Operating Environment

The system is delivered as a web-based application, accessible through standard modern browsers and It integrates directly with Microsoft SharePoint, to which it publishes approved documents as the current controlled version once the approval workflow is complete.

Design and Implementation Constraints

- The system must validate documents against a defined ISO document format/template; documents that do not conform must be rejected at upload with a clear error message, and the user must be returned to the upload step.
- The approver for any given document must be determined programmatically from an approval matrix rather than manually selected by the uploader, to ensure consistent governance.
- Only one active/published version of a given document may exist in SharePoint at a time; new versions must supersede prior versions in a traceable manner.

Assumptions

- It is assumed that only one approver (or a defined approval chain) is required per document, as determined by the matrix; multi-level or parallel approval is not addressed unless explicitly required.
- It is assumed that the approval matrix (mapping document types/attributes to approvers) is maintained and kept up to date by administrators

System Features and Requirements

Functional Requirements

- Upload Document - The system shall allow users to upload documents
- Validate Document Format - The system shall validate the document if invalid send error message and if valid The system shall verify compliance with ISO document structure.
- Duplicate Document Detection - The system shall determine whether the uploaded document already exist, if the document does not exist,the system shall continue with the approval workflow otherwise initiate document comparison.
- Document comparison - The system shall compare uploaded document with existing document if identify any added, deleted or modified section created new version and if not notify already exist.
- Version management - when changes are detected, the system shall create a new version
- Approval Workflow - The system shall determine the appropriate approver based on the approval matrix.
- Document Approval - The approver shall review document details and approve or reject, if reject add a comment
- User Notification - The system shall notify document creators of approval or reject decisions.
- Publish Document - The system shall publish approved documents to SharePoint.

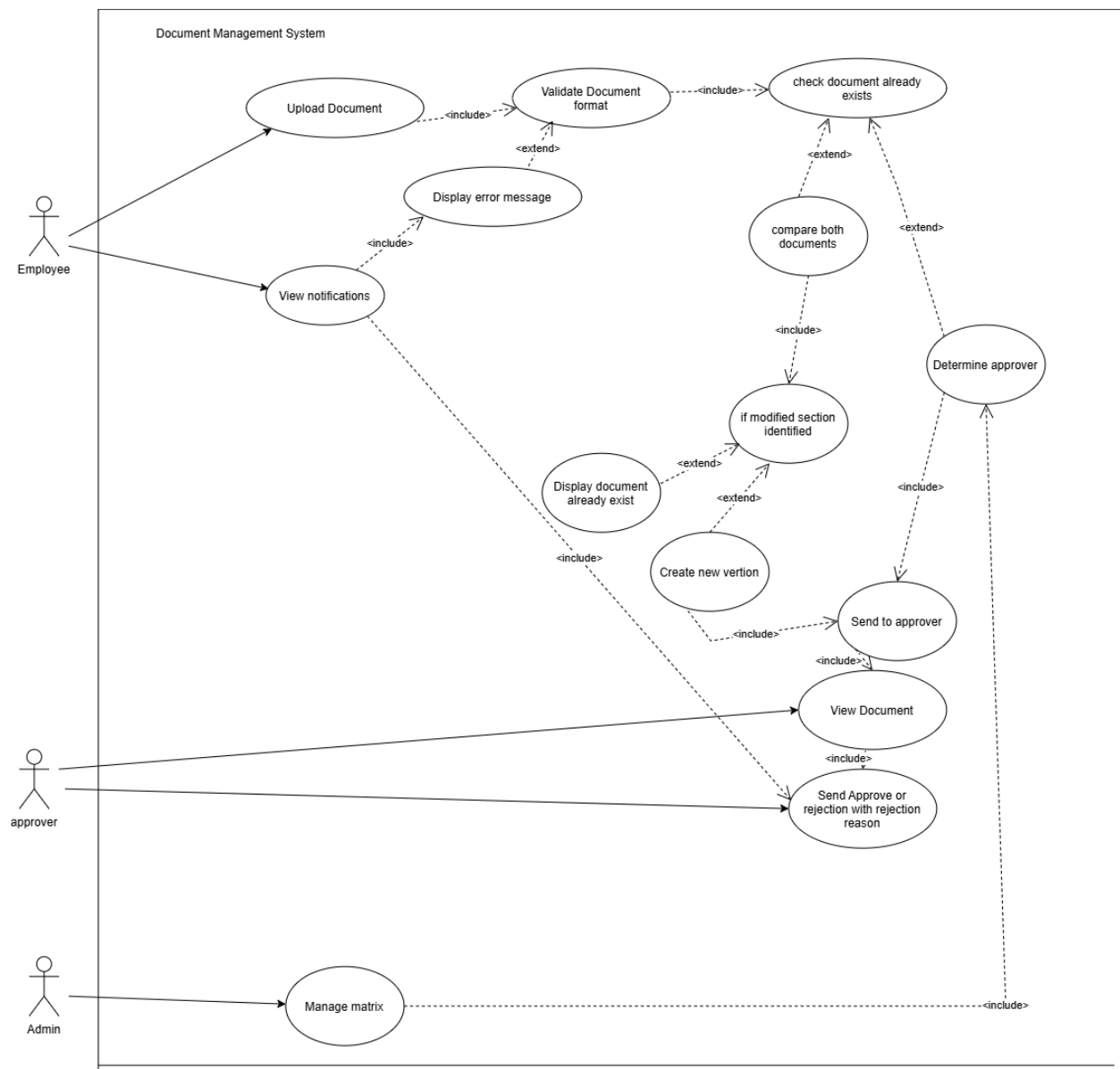
Non – Functional Requirements

- Performance - The system shall validate document within 5 second and support multiple users simultaneously
- Security - Only authorized users shall access the system give role-based access control.
- Reliability - The system shall provide 99% availability during working hours.
- Usability - The user interface shall be simple and easy to use
- Scalability - The system shall support an increasing number of users and document.

Database Requirements

- The system shall store user information, including user ID, name, email, and role.
- The system shall store document details such as document ID, title, version, upload date, status, and uploader.
- The system shall store approval workflow details, including approver, approval status, comments, and approval date.
- The system shall maintain audit logs for uploads, approvals, rejections, and publishing activities.

Use case Diagram



Business Logics

1. Upload & format validation

- A document must conform to the organization's ISO template (heading structure, metadata fields, numbering) to be accepted.
- If validation fails → reject the upload, display "Invalid format," and return the user to the upload screen. No document record is created for failed uploads.
- If validation passes → proceed to the duplicate/existing-document check.

2. Duplicate / existing document check

- The system checks whether a document with the same identity already exists — typically matched by document title + document type + department, or an internal document code if one is assigned at creation.
- **If no match exists** → treat as a brand-new document. Skip comparison, go directly to approver determination.
- **If a match exists** → do not create a new document record yet. Instead, pull the latest existing version and proceed to comparison.

3. Version comparison

- Compare the newly uploaded file against the current active version, section by section (not just a byte-level diff).
- Classify differences as: added, removed, or modified sections.
- **If no differences are found** → the upload is functionally identical to what's already published. Notify the user "document already exists" and stop the workflow — no new version is created, no approval is triggered.
- **If differences are found** → increment the version number and create a new version record. The prior version is retained, not overwritten, for audit purposes.

4. Approver determination

- The approver is never chosen manually by the uploader — it's derived automatically from the **approval matrix**, keyed on attributes like document type, department, and possibly criticality/risk level.

5. Approval request & decision

- An approval request is sent to the determined approver only
- **If rejected** → notify the originator, close out that version's workflow. The document stays at its last-approved published version (if one exists); the rejected version is retained for audit but never published.
- **If approved** → notify the originator, then proceed to publish.
- A rejected version, if resubmitted, should probably be treated as a new upload rather than resuming — worth clarifying whether corrections create yet another new version or amend the rejected one.

6. Publishing

- Only an approved version can be published to SharePoint.

Tables and fields with data type

User

Field	Data type
user_id(PK)	INT
name	VARCHAR(100)
email	VARCHAR(150)
role	VARCHAR(50)

Document

Field	Data type
document_id(PK)	INT
title	VARCHAR(255)
file_type	VARCHAR(255)
status	VARCHAR(255)
uploaded_by(FK)	VARCHAR(255)
current_version	VARCHAR(255)
sharepoint_link	VARCHAR(255)

Approval

Field	Data type
approval_id(PK)	INT
decision	VARCHAR(20)
decision_date	DATETIME
approval_by(FK)	INT
document_id(FK)	INT
matrix_id(FK)	INT

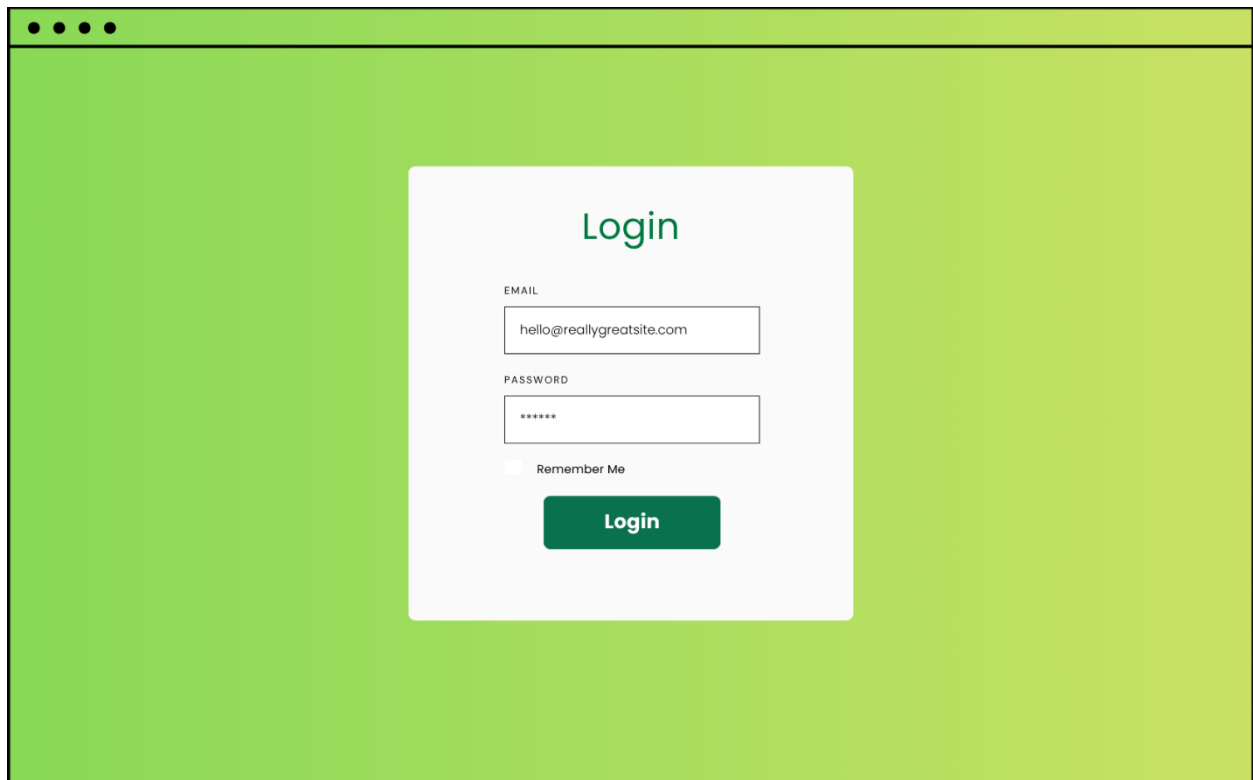
Approval_matrix

Field	Data type
matrix_id(PK)	INT
document_type	VARCHAR(50)
department	VARCHAR(50)
approval_name	VARCHAR(50)

Notification

Field	Data type
notification_id(PK)	INT
notification_type	VARCHAR(50)
notification_description	VARCHAR(500)
date	DATETIME
document_id(FK)	INT

User Interface



The image shows a login page with a green background. In the center, there is a white rounded rectangle containing the login form. The form has a title 'Login' in green. Below the title, there are two input fields: 'EMAIL' with the value 'hello@reallygreatsite.com' and 'PASSWORD' with the value '*****'. Below the password field, there is a checkbox labeled 'Remember Me'. At the bottom of the form, there is a green button with the text 'Login'.

Figure 1 Login page

Upload Document

Description

Department

Clear

Upload

Figure 2 upload Document

My Documents

Search

+ New

Doc ID	Document	Status	Upload Date	Approval Date	Details
00102	xyz	Approved	2026/08/01	2026/08/08	->
00101	ABC	Rejected	2026/07/30	2026/08/08	->

Figure 3 User document(history)

My Documents

Search

Document ID

Document

DOC

xyz

Upload Date

Approval Date

Approval by

Department

Sharepoint link

Status

Description

Comments

Download

Figure 4 Full Document Details

Approval Request

Request By

Document ID

Document

DOC

Comments

Description

✓ Document In ISO format

Already Exist

✓

✗

To find motified section

Use AI

Reject

Approve

Figure 5 Approval Request

ER Diagram

